Lab No: 06(

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Semester 2nd

Section B

Registration BF25NWELE0686

Lab Title: For Loop in C++

Learning objectives:

The following topics will be covered in this lab session

- For Loop
- Nested For Loop

For Loop:

A for loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times.

Syntax:

The syntax of a for loop in C++ is:

```
for ( init; condition; increment )
{
    statement(s);
}
```

Here is the flow of control in a for loop:

- The **initial** step is executed first, and only once. This step allows you to declare and initialize any loop control variables. You are not required to put a statement here, as long as a semicolon appears.

- Next, the **condition** is evaluated. If it is true, the body of the loop is executed. If it is false, the body of the loop does not execute and flow of control jumps to the next statement just after the for loop.
- After the body of the for loop executes, the flow of control jumps back up to the **increment** statement. This statement allows you to update any loop control variables. This statement can be left blank, as long as a semicolon appears after the condition.
- The condition is now evaluated again. If it is true, the loop executes and the process repeats itself (body of loop, then increment step, and then again condition). After the condition becomes false, the for loop terminates.

Example:

Write a program using for loop which prints the values from 10 to 1.

```
//This program will print 10 to 1 using for loop

#include <iostream.h>

using namespace std;

int main()
{
        for (inti=10;i>0;i--)
        {
          cout<<i<<"\n";
        }

        cout<<"outside loop now";

}
```

➴ Example:

Write a program that checks if a number is prime or not.

```
//checksifanumberisprime
#include <iostream.h>
#include <conio.h>
Int  main()
{
        intnum;
        cout<<"Enter a number please\n";
        cin>>num; int flag=0;
        for (inti=2;i<num;i++)
        {
          if (num%i==0)
                {
                  flag=1; break;
                }
}
        if (flag==1)
        {
           cout<<"it is not a prime number\n";
        }
        else
        {
          cout<<"it is a prime number\n";
        }
}
}
```

➴ Nested For Loop:

One `for` loop is placed **inside the body of another `for` loop**.

How it works

- The **outer loop** runs first
- For **each iteration** of the outer loop, the **inner loop runs completely**

☞ So the inner loop repeats again and again for every outer loop cycle

General Syntax

```
for(initialization1; condition1; update1)
{
   for(initialization2; condition2; update2)
   {
      // code to execute
   }
}
```

Example Program

```
#include <iostream>
using namespace std;

int main()
{
   for(int i = 1; i <= 3; i++)   // Outer loop
   {
      for(int j = 1; j <= 2; j++)  // Inner loop
      {
         cout << "i = " << i << ", j = " << j << endl;
      }
   }

   return 0;
}
```

Output

```
i = 1, j = 1
i = 1, j = 2
i = 2, j = 1
i = 2, j = 2
i = 3, j = 1
i = 3, j = 2
```

⭑ **Simple Example (Pattern)**

```
for(int i = 1; i <= 3; i++)
{
   for(int j = 1; j <= 3; j++)
   {
      cout << "* ";
   }
   cout << endl;
}
```

➯ Lab tasks:

Task :1

• Write a C++ program that uses a while loop to display numbers from 1 to 10.
• Display each number on a separate line.

```
#include <iostream>                         // Include input-output library

using namespace std;                        // Use standard namespace

int main()                                  // Main function where execution starts{

int i = 1;                                  // Initialize variable i with value 1

while (i <= 10){                            // Loop runs as long as i is less than or equal to 10

   cout << i << endl;                       // Print current value of i on a new line

      i++;                                  // Increment i by 1 after each iteration

   }

return 0;                                   // End of program

}
```

 Output: 1 to 10 , each in a separate line

Task 2:

Nested *for* Loop : Multiplication Table

• Write a C++ program that generates a multiplication table from 1 to 10.

• Use nested *for* loops to iterate through the rows and columns of the table.

```
#include <iostream>                                 // Library for input-output

using namespace std;                                // Use standard namespace

int main()                                          // Main function

{

  for(int i = 1; i <= 10; i++)                      // Outer loop for rows (numbers 1 to 10)

  {

    for(int j = 1; j <= 10; j++)                    // Inner loop for columns (multipliers 1 to 10)

    {

                                                    // Print product of i and j with some spacing

      cout << i * j << "\t";                        // \t adds tab space for alignment

    }


    cout << endl;                                   // Move to next line after each row

  }

  return 0;                 // End of program

}
```

🔍 **Sample Output**

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30

Task 3: Sum using For Loop:

- Write a C program that prompts the user to enter a positive integer.
- Use a for loop to calculate the sum of all numbers from 1 to the entered integer.
- Display the calculated sum

```
#include <iostream>                          // Include input-output stream library
using namespace std;                         // Use standard namespace

int main()                                   // Main function where execution starts
{
    int num;                                 // Variable to store user input
    int sum = 0;                             // Variable to store sum, initialized to 0

    cout << "Enter a positive integer: ";    // Prompt user
    cin >> num;                              // Take input from user

                                             // For loop to calculate sum from 1 to num
    for(int i = 1; i <= num; i++)            // Loop runs from 1 to num
    {
        sum = sum + i;                       // Add current value of i to sum
    }
    cout << "Sum from 1 to " << num << " is: " << sum << endl;     // Display the result

    return 0;                                // End of program
}
```

Explanation (Simple)

- User enters a number (e.g., **4**)
- Loop runs like:

```
sum = 1 + 2 + 3 + 4 = 10
```

Task 4: Nested For Loop - Pattern Printing

- Write a C program that uses nested for loops to print a pattern of asterisks.
- The pattern should form a triangle shape, with each row having an increasing number of asterisks.
- The number of rows in the pattern should be determined by user input.

```
#include <iostream>                                  // Library for input-output
using namespace std;                                 // Use standard namespace

int main()                                           // Main function
{
   int rows;                                         // Variable to store number of rows

   cout << "Enter number of rows: ";                 // Ask user for input
   cin >> rows;                                      // Take input from user

    for(int i = 1; i <= rows; i++)                   // Outer loop controls number of rows

   {
       for(int j = 1; j <= i; j++)                   // Inner loop prints stars in each row

     {
        cout << "* ";                                // Print star with space
     }

     cout << endl;                                   // Move to next line after each row
   }

   return 0;                                         // End of program
}
```

How it works (Simple Explanation)

- Outer loop (`i`) → controls **rows**
- Inner loop (`j`) → prints **stars**
- Each row prints stars equal to its row number

Example Output (rows = 5)

```
*

* *

* * *

* * * *

* * * * *
```